8.033 Fall 2025: Solution Set 7

1

Using the definition of the Λ matrix, we show

$$\begin{aligned}
\Lambda[\varphi_1]\Lambda[\varphi_2] &= \begin{bmatrix} \cosh\varphi_1 & -\sinh\varphi_1 \\ -\sinh\varphi_1 & \cosh\varphi_1 \end{bmatrix} \begin{bmatrix} \cosh\varphi_2 & -\sinh\varphi_2 \\ -\sinh\varphi_2 & \cosh\varphi_2 \end{bmatrix} \\
&= \begin{bmatrix} \cosh\varphi_1\cosh\varphi_2 + \sinh\varphi_1\sinh\varphi_2 & -\cosh\varphi_1\sinh\varphi_2 - \sinh\varphi_1\cosh\varphi_2 \\ -\sinh\varphi_1\cosh\varphi_2 - \cosh\varphi_1\sinh\varphi_2 & \sinh\varphi_1\sinh\varphi_2 + \cosh\varphi_1\cosh\varphi_2 \end{bmatrix} \\
&= \begin{bmatrix} \cosh(\varphi_1+\varphi_2) & -\sinh(\varphi_1+\varphi_2) \\ -\sinh(\varphi_1+\varphi_2) & \cosh(\varphi_1+\varphi_2) \end{bmatrix},
\end{aligned}$$

where we have used the angle addition formula for hyperbolic functions to get the last line. We recognize that the last line of Eq. (22) is just $\Lambda[\varphi_1+\varphi_2]$, so we have shown

$$\Lambda[\varphi_1]\Lambda[\varphi_2] = \Lambda[\varphi_1+\varphi_2].$$

2

(a) $U^aV^bg_{ab}$ is a Lorentz scalar, so it is invariant under coordinate transformations. In other words, $U^aV^bg_{ab}$ computed using the components of U^a, V^b, and g_{ab} in a frame R should agree with $U^{a'}V^{b'}g_{a'b'}$ computed in a different frame R'. We can check this explicitly by Lorentz transforming the components of U^a, V^b, and g_{ab} using the $\Lambda^\mu{}_\nu$

$$U^{\mu'}V^{\nu'}g_{\mu'\nu'} = (\Lambda^{\mu'}{}_\mu U^\mu)(\Lambda^{\nu'}{}_\nu V^\nu)g_{\mu'\nu'} = U^\mu V^\nu(g_{\mu'\nu'}\Lambda^{\mu'}{}_\mu\Lambda^{\nu'}{}_\nu) = U^\mu V^\nu g_{\mu\nu}.$$

(b) Let $\{x^\mu\}$ be an arbitrary set of coordinates. Now consider $\partial x^\mu/\partial x^\nu$. If $\mu=\nu$, then x^μ and x^ν are the same variable, so we have

$$\frac{\partial x^\mu}{\partial x^\nu} = 1, \text{ if } \mu=\nu.$$

If $\mu\neq\nu$, then x^μ and x^ν are independent variables, so we have

$$\frac{\partial x^\mu}{\partial x^\nu} = 0, \text{ if } \mu\neq\nu.$$

Combining Eqs. (25) and (26), we get

$$\frac{\partial x^\mu}{\partial x^\nu} = \delta^\mu{}_\nu.$$

(c) In class, we have learned that we can use $\Lambda^\mu{}_{\mu'}$ to Lorentz transform coordinates from the primed frame to the unprimed frame (both in Cartesian coordinates), namely

$$x^\mu = \Lambda^\mu{}_{\mu'}x^{\mu'}.$$

Then we can compute

$$\frac{\partial x^\mu}{\partial x^{\mu'}} = \frac{\partial(\Lambda^\mu{}_{\nu'}x^{\nu'})}{\partial x^{\mu'}} = \Lambda^\mu{}_{\nu'}\frac{\partial x^{\nu'}}{\partial x^{\mu'}} = \Lambda^\mu{}_{\nu'}\delta^{\nu'}{}_{\mu'} = \Lambda^\mu{}_{\mu'}.$$

3 Acceleration 4- and 3-vectors

(a) Let's choose our coordinates such that $\vec{a}$ and $\vec{v}$ are along the x-axis. Then we just need to consider A^0 and A^1 since $A^2 = A^3 = 0$. By definition, we have

$$A^0 = \frac{dU^0}{d\tau} = \frac{d\gamma}{dt}\frac{dt}{d\tau} = \gamma\frac{d\gamma}{dt}, \tag{1}$$

where we have used $U^0 = \gamma$ and used the chain rule to get the second equality, and used $dt/d\tau = \gamma$ to get the third equality in Eq. (1). Now use $\gamma = 1/\sqrt{1-v^2}$, where v is the speed of the object in our coordinate system, and plug into Eq. (1), we get

$$A^0 = \frac{1}{\sqrt{1-v^2}}\frac{d}{dt}\left(\frac{1}{\sqrt{1-v^2}}\right) = \frac{v}{(1-v^2)^2}\frac{dv}{dt} = \gamma^4 va. \tag{2}$$

And, similarly

$$A^1 = \frac{dU^1}{d\tau} = \frac{d(\gamma v)}{d\tau} = v\frac{d\gamma}{d\tau} + \gamma\frac{dv}{d\tau} = vA^0 + \gamma\frac{dv}{dt}\frac{dt}{d\tau} = \gamma^4 v^2 a + \gamma^2 a = \gamma^4 a. \tag{3}$$

(b) Let the primed coordinates be those of the MCRF. In the MCRF, the speed of the object $u = 0$ at the instance, although $du/dt \neq 0$ and is along the x-axis by our choice of the spatial axes. By definition,

$$A^{0'} = \frac{dU^{0'}}{d\tau} = \frac{d\gamma(u)}{dt} = \frac{d\gamma(u)}{du}\frac{du}{dt} = 0, \tag{4}$$

where we have used the fact $dt = d\tau$ in a MCRF, and the last equality gives zero because taking the derivative of $\gamma(u)$ with respect to u gives a prefactor of u which is zero.

Similarly,

$$A^{1'} = \frac{dU^{1'}}{d\tau} = \frac{d[\gamma(u)u]}{dt} = \gamma(u)\frac{du}{dt} = \frac{du}{dt} = a'. \tag{5}$$

(c) We have just computed the components of the acceleration 4-vector in an arbitrary frame and the MCRF. Since the acceleration 4-vector is a 4-vector, we know its components in the two different frames are related by the Lorentz transform

$$\begin{bmatrix} A^0 \\ A^1 \end{bmatrix} = \begin{bmatrix} \gamma & \gamma v \\ \gamma v & \gamma \end{bmatrix} \begin{bmatrix} A^{0'} \\ A^{1'} \end{bmatrix}. \tag{6}$$

Using the values of A^μ and $A^{\mu'}$ we have computed in the previous parts, Eq. (6) gives two equations

$$\gamma^4 va = \gamma va', \tag{7}$$
$$\gamma^4 a = \gamma a'. \tag{8}$$

Both equations suggest that

$$a = \gamma^{-3} a'. \tag{9}$$

(d) Suppose a' is a constant (i.e., the object is undergoing a constant acceleration), then Eq. (9) tells us that

$$a = \frac{dv}{dt} = a'(1 - v^2)^{3/2}, \tag{10}$$

or equivalently

$$a' dt = \frac{dv}{(1 - v^2)^{3/2}}, \tag{11}$$

which we can integrate to solve for v as a function of t:

$$t = \frac{1}{a'} \int \frac{dv}{(1 - v^2)^{3/2}}, \tag{12}$$

which can be done using a trig substitution (e.g., $v = \sin(w)$). After setting $v(t = 0) = 0$ and rearranging the equation, we get

$$v = \frac{dx}{dt} = \frac{a't}{\sqrt{1 + (a')^2 t^2}}. \tag{13}$$

Then we can integrate again to get the position x as a function of time. Setting $x(t = 0) = 0$, we get

$$x = \frac{1}{a'}(\sqrt{1 + (a')^2 t^2} - 1). \tag{14}$$

Note that we could also set $x(t = 0) = 1/a'$, in that case we get

$$x = \frac{1}{a'} \sqrt{1 + (a')^2 t^2}, \tag{15}$$

and equivalently,

$$x^2 - t^2 = \frac{1}{(a')^2}, \tag{16}$$

which is to say the trajectory is a hyperbola with a fixed invariant distance of $1/a'$ from the origin, consistent with what we have seen in the lecture. The coordinates of the object along the trajectory is simply $(t, x) = (t, \frac{1}{a'}\sqrt{1 + (a')^2 t^2})$.

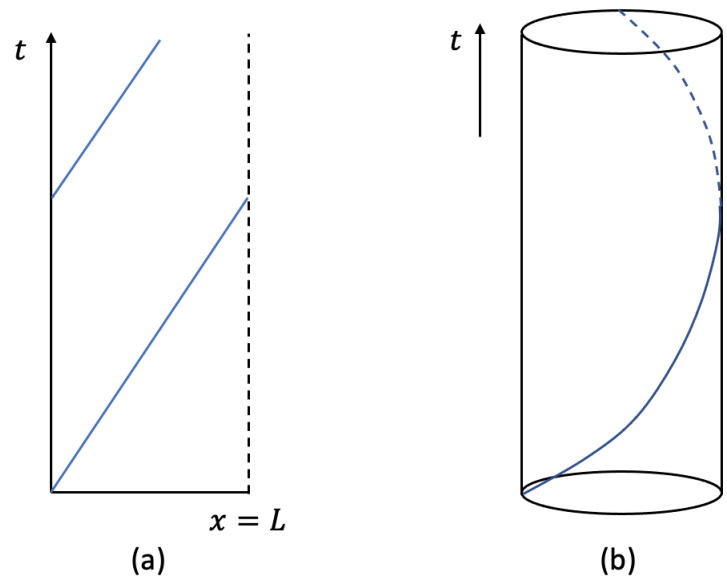

Figure 1: Panel (a): Barbara's worldline in Alice's frame on a standard spacetime diagram. Barbara has already passed by Alice once. Panel (b): Barbara's worldline in Alice's frame drawn on a cylinder, not to the same scale as panel (a). In this diagram, Barbara has yet to make a full trip around.

4 Twin paradox in a compact universe

(a) See Fig. (1). Comment: students only need to show one of the two diagrams.

(b) In Alice's frame, Barbara needs to travel a total distance of L to make her way back. Since Alice measures Barbara to travel at speed v, the time elapsed during the trip measured by Alice is simply

$$\Delta t_A = \frac{L}{v}. \tag{17}$$

Due to time dilation, Alice expects Barbara to measure

$$\Delta t_B = \frac{\Delta t_A}{\gamma} = \frac{L}{\gamma v}. \tag{18}$$

(c) In Alice's frame, the periodic identification is

$$(t, x, y, z) \sim (t, x + L, y, z). \tag{19}$$

To find the periodic identification in Barbara's frame, we need to perform a Lorentz transform on (t, x, y, z)

$$\begin{aligned} \tilde{t} &= \gamma(t - vx), \\ \tilde{x} &= \gamma(x - vt), \\ \tilde{y} &= y, \\ \tilde{z} &= z, \end{aligned} \tag{20}$$

where the coordinates with tildes are the ones in Barbara's frame, and perform a Lorentz transform on $(t, x+L, y, z)$

$$\begin{aligned}
\tilde{t}' &= \gamma[t - v(x+L)] = \tilde{t} - \gamma v L, \\
\tilde{x}' &= \gamma[(x+L) - vt] = \tilde{x} + \gamma L, \\
\tilde{y}' &= y, \\
\tilde{z}' &= z.
\end{aligned} \tag{21}$$

We therefore find the periodic identification in Barbara's frame is

$$(\tilde{t}, \tilde{x}, \tilde{y}, \tilde{z}) \sim (\tilde{t} - \gamma v L, \tilde{x} + \gamma L, \tilde{y}, \tilde{z}). \tag{22}$$

In Barbara's frame, Alice started moving from $\tilde{x} = 0$ with the speed v in the $-x$ direction. When Alice reached $\tilde{x} = -\gamma L$, her coordinates in Barbara's frame are

$$\left(\frac{\gamma L}{v}, -\gamma L, 0, 0\right), \tag{23}$$

where we are assuming $y = \tilde{y} = 0$ and $z = \tilde{z} = 0$. Using Barbara's periodic identification, this is the same as

$$\left(\frac{\gamma L}{v} - \gamma v L, 0, 0, 0\right). \tag{24}$$

In other words, Alice now has returned to Barbara, and Barbara's clock measures the time elapsed to be

$$\Delta\tilde{t} = \frac{\gamma L}{v} - \gamma v L = \frac{L}{\gamma v} = \Delta t_B. \tag{25}$$

(d) The length of the compact direction (x-axis) in Alice's frame is L, while it is γL in any other inertial observer's frame. Alice can compare the length of her compact direction with those of others. She then concludes that her frame is special because the length of the compact direction in her frame is minimal. This is in no violation of the fact that no coordinate system in Minkowski space is special because the periodic identification breaks the translation symmetry of the Minkowski space.

5

We can construct a counterexample, namely showing that we can find a spacelike curve connecting two timelike-separated events p and q. Consider a case in two spatial dimensions plus a time axis. We can easily construct a spacelike curve between two events conneted by a timelike worldline. See Fig. (2).

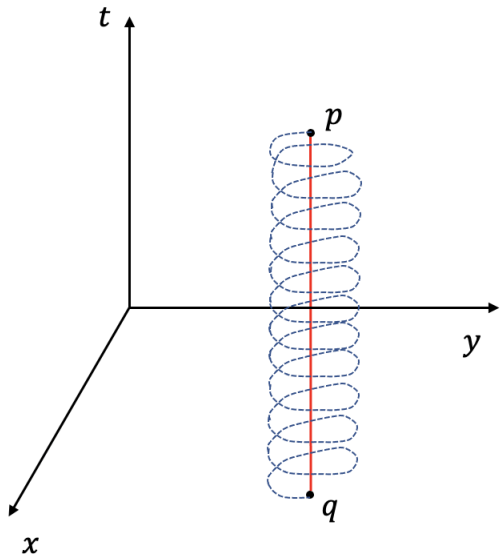

Figure 2: The red solid line is a timelike worldline connecting events p and q. The blue dashed curve is a spacelike curve that connects p and q. We have thus found a counterexample to the statement that p and q are spacelike-separated if there exists a spacelike curve between them. Here p and q are connected by a spacelike curve, but p and q are timelike separated.